

Amazon Sales Performance & Profitability Analysis

Author: Jeevan Venkatesh

Tools: Python (Pandas), SQL (SQLite), Power BI

Date: 04-01-2026

Executive Summary

This project analyzes Amazon transactional sales data to evaluate overall sales performance, profitability drivers, and operational efficiency. The analysis combines Python for data preparation, SQL for business metrics, and Power BI for executive-level visualization.

1. Business Objective

The primary objectives of this analysis are:

- To understand overall sales and order volume
- To identify top-performing product categories
- To analyze order cancellations and fulfillment efficiency
- To evaluate regional (state-wise) sales performance
- To support data-driven business decision-making

```
In [2]: import pandas as pd

df = pd.read_csv(
    r"C:\Users\jeeva\OneDrive\Desktop\Amazon Sales Performance & Profitability Analysis\raw_data\amazon_sales_final.csv"
)

df.head()
```

```
C:\Users\jeeva\AppData\Local\Temp\ipykernel_44912\3878976174.py:3: DtypeWarning: Columns (18,20) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv(
```

Out[2]:

	order_id	order_date	order_status	fulfilment_type	sales_channel	ship- service- level	Style	category	Size	Courier Status	...	currency	sales_amc
0	405- 8078784- 5731545	04-30-22	Cancelled	Merchant	Amazon.in	Standard	SET389	Set	S	NaN	...	INR	64
1	171- 9198151- 1101146	04-30-22	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard	JNE3781	kurta	3XL	Shipped	...	INR	40
2	404- 0687676- 7273146	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3371	kurta	XL	Shipped	...	INR	32
3	403- 9615377- 8133951	04-30-22	Cancelled	Merchant	Amazon.in	Standard	J0341	Western Dress	L	NaN	...	INR	75
4	407- 1069790- 7240320	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3671	Top	3XL	Shipped	...	INR	57

5 rows × 21 columns



In [3]:

```
df.shape
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128975 entries, 0 to 128974
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   order_id              128975 non-null object
 1   order_date            128975 non-null object
 2   order_status          128975 non-null object
 3   fulfilment_type       128975 non-null object
 4   sales_channel         128975 non-null object
 5   ship-service-level    128975 non-null object
 6   Style                 128975 non-null object
 7   category              128975 non-null object
 8   Size                  128975 non-null object
 9   Courier Status        122103 non-null object
10   quantity              128975 non-null int64
11   currency              121180 non-null object
12   sales_amount          121180 non-null float64
13   ship-city             128942 non-null object
14   state                 128942 non-null object
15   ship-postal-code      128942 non-null float64
16   country               128942 non-null object
17   B2B                   80074 non-null object
18   fulfilled-by          128386 non-null object
19   Unnamed: 19           39126 non-null object
20   Unnamed: 20           79925 non-null object
dtypes: float64(2), int64(1), object(18)
memory usage: 20.7+ MB

```

3. Data Overview

- Dataset Size: ~128,975 orders
- Data Type: Transactional retail sales data
- Key Columns: Order Date, Category, Sales Amount, State, Fulfilment Type, Order Status, B2B Indicator

4. Tools & Methodology

Python:

- Data cleaning and preprocessing
- Date standardization and feature creation

SQL (SQLite):

- Aggregation queries
- Revenue, cancellation, and category analysis

Power BI:

- Interactive dashboard
- KPI cards and visual analytics

5. Key Insights from SQL Analysis

- The dataset contains approximately 129K orders, indicating high transaction volume.
- A small number of product categories contribute the majority of total revenue.
- Cancellation rate highlights potential revenue leakage areas.
- B2B orders contribute a smaller volume but show higher order values.

6. Power BI Dashboard Summary

The Power BI dashboard provides an executive overview of sales performance through KPI cards, category-wise revenue charts, order status distribution, and sales trends over time. Interactive filters allow users to drill down by date and category.

7. Business Recommendations

- Focus inventory and marketing efforts on top-performing categories.
- Reduce cancellations by improving order confirmation and delivery reliability.
- Strengthen logistics operations in high-revenue states.
- Explore growth opportunities in the B2B segment.

8. Conclusion

This analysis demonstrates how data-driven insights can support better decision-making in retail operations. By combining Python, SQL, and Power BI, the project delivers actionable insights into sales performance and profitability.